

Table S1 Plantation estates selected in the Midlands and Zululand. The location of each plantation estate is given, along with the number of indigenous forest (For), grassland (Grass) and eucalyptus (Euc) sites within each estate.

Region	Plantation Estates	Plantation location	For	Grass	Euc
Midlands	Batchelors	30°09'12.77"S, 29°47'53.37"E	2	1	4
	Cottingham	30°07'23.7"S, 29°47'14.49"E	1		
	Fabershill	29°40'13.3"S, 29°55'59.5"E		5	
	Harcourt	29°51'00.2"S, 30°10'36.1"E			5
	Maybole	30°13'53.64"S, 29°44'8.95"E	12	7	3
	Mkhuzane	29°49'33.9"S, 30°18'10.8"E			6
	Mount Shannon	29°41'11.8"S, 29°58'43.0"E	3	5	
Region	Plantation Estates	Plantation location	For	Grass	Euc
Zululand	Canewood	32°05'42.84"S, 28°37'56.52"E	3	2	2
	Mcilrath	31°58'38.14"S, 28°42' 53.61"E	2	3	2
	Montigny	32°11'9.12"S, 28°35'41.9"E	1	2	1
	Port Dunford	31°50'16.7"S, 28°53'30.92"E	5	6	6
	SQF Kwambonambi	32°10'27.06"S, 28°39'26.68"E	7	5	7

Table S2 Species list and trait identities used in the analysis of ants in the Midlands and Zululand.

Subfamily	Genus	Forage strategy	Habitat position	Eye Size	Body length (mm)	Femur length (mm)	Colour value (V)
Dolichoderinae	<i>Tapinoma</i> sp. 1	GenForager	Litter	Large	1.50	0.50	7
Dolichoderinae	<i>Tapinoma</i> sp. 2	GenForager	Litter	Large	1.00	1.00	31
Dolichoderinae	<i>Tapinoma</i> sp. 3	GenForager	Litter	Large	1.00	0.50	36
Dolichoderinae	<i>Tapinoma</i> sp. 4	GenForager	Litter	Large	1.00	1.00	24
Dorylinae	<i>Aenictus</i> sp. 1	Pred	Soil	Absent	0.50	0.25	38
Dorylinae	<i>Aenictus</i> sp. 2	Pred	Soil	Absent	1.00	0.50	65
Dorylinae	<i>Aenictus</i> sp. 3	Pred	Soil	Absent	1.50	1.00	58
Formicinae	<i>Anoplolepis</i> sp. 1	GenForager	Litter	Large	0.50	0.50	62
Formicinae	<i>Anoplolepis</i> sp. 2	GenForager	Litter	Large	0.50	0.25	54
Formicinae	<i>Camponotus</i> sp. 1	GenForager	Soil	Moderate	3.50	5.00	2
Formicinae	<i>Camponotus</i> sp. 2	GenForager	Soil	Moderate	2.00	2.00	15
Formicinae	<i>Camponotus</i> sp. 3	GenForager	Soil	Moderate	3.00	2.00	49
Formicinae	<i>Camponotus</i> sp. 4	GenForager	Soil	Moderate	1.50	1.50	27
Formicinae	<i>Camponotus</i> sp. 5	GenForager	Soil	Moderate	2.50	2.00	53
Formicinae	<i>Camponotus</i> sp. 6	GenForager	Soil	Moderate	2.00	1.50	50
Formicinae	<i>Camponotus</i> sp. 7	GenForager	Soil	Moderate	2.00	1.50	15
Formicinae	<i>Camponotus</i> sp. 8	GenForager	Soil	Moderate	2.00	2.00	48
Formicinae	<i>Camponotus</i> sp. 9	GenForager	Soil	Moderate	2.00	1.50	26
Formicinae	<i>Camponotus</i> sp. 10	GenForager	Soil	Moderate	3.00	3.00	50
Formicinae	<i>Camponotus</i> sp. 11	GenForager	Soil	Moderate	4.00	2.00	33
Formicinae	<i>Camponotus</i> sp. 12	GenForager	Soil	Moderate	2.50	1.50	29
Formicinae	<i>Camponotus</i> sp. 13	GenForager	Soil	Moderate	2.00	2.00	66
Formicinae	<i>Camponotus</i> sp. 14	GenForager	Soil	Moderate	3.00	2.00	71
Formicinae	<i>Camponotus</i>	GenForager	Soil	Moderate	3.00	2.00	25

	sp. 15						
Formicinae	<i>Camponotus</i> sp. 16	GenForager	Soil	Moderate	3.00	2.00	19
Formicinae	<i>Lepisiota</i> sp. 1	GenForager	Soil	Moderate	1.00	0.50	32
Formicinae	<i>Lepisiota</i> sp. 2	GenForager	Soil	Moderate	0.50	0.50	33
Formicinae	<i>Lepisiota</i> sp. 3	GenForager	Soil	Moderate	1.00	0.50	53
Formicinae	<i>Lepisiota</i> sp. 4	GenForager	Soil	Moderate	0.50	0.50	28
Formicinae	<i>Lepisiota</i> sp. 5	GenForager	Soil	Moderate	1.50	0.50	30
Formicinae	<i>Lepisiota</i> sp. 6	GenForager	Soil	Moderate	0.50	0.50	25
Formicinae	<i>Lepisiota</i> sp. 7	GenForager	Soil	Moderate	1.00	0.50	14
Formicinae	<i>Lepisiota</i> sp. 8	GenForager	Soil	Moderate	2.00	1.00	36
Formicinae	<i>Lepisiota</i> sp. 9	GenForager	Soil	Moderate	0.10	0.50	18
Formicinae	<i>Lepisiota</i> sp. 10	GenForager	Soil	Moderate	1.50	1.50	32
Formicinae	<i>Paratrechina</i> sp. 1	GenForager	Soil	Large	1.00	1.00	29
Formicinae	<i>Paratrechina</i> sp. 2	GenForager	Soil	Large	1.00	0.50	35
Formicinae	<i>Paratrechina</i> sp. 3	GenForager	Soil	Large	1.00	1.00	70
Formicinae	<i>Paratrechina</i> sp. 4	GenForager	Soil	Large	2.50	1.50	65
Formicinae	<i>Plagiolepis</i> sp. 1	Sap	Litter	Large	0.50	0.25	30
Formicinae	<i>Plagiolepis</i> sp. 2	Sap	Litter	Large	1.00	0.50	45
Formicinae	<i>Plagiolepis</i> sp. 3	Sap	Litter	Large	1.00	0.50	65
Formicinae	<i>Plagiolepis</i> sp. 4	Sap	Litter	Large	0.50	0.30	68
Formicinae	<i>Polyrhachis</i> sp. 1	GenForager	Soil	Large	3.00	2.50	25
Formicinae	<i>Polyrhachis</i> sp. 2	GenForager	Soil	Large	3.00	2.50	28
Formicinae	<i>Polyrhachis</i> sp. 3	GenForager	Soil	Large	3.00	2.50	26
Ponerinae	<i>Anochetus</i> sp. 1	Pred	Litter	Large	1.00	1.00	63
Ponerinae	<i>Boloponera</i> sp. 1	Pred	Litter	Absent	1.00	1.00	48
Ponerinae	<i>Bothroponera</i> sp. 1	Pred	Litter	Small	1.50	1.00	2
Ponerinae	<i>Bothroponera</i> sp. 2	Pred	Litter	Small	5.00	4.00	35
Ponerinae	<i>Bothroponera</i> sp. 3	Pred	Litter	Small	3.00	2.50	24
Ponerinae	<i>Bothroponera</i> sp. 4	Pred	Litter	Small	2.50	2.00	22

Ponerinae	<i>Bothroponera</i> sp. 5	Pred	Litter	Small	2.50	2.50	21
Myrmicinae	<i>Cataulacus</i> sp. 1	Sap	Veg	Large	1.50	0.50	16
Myrmicinae	<i>Crematogaster</i> sp. 1	GenForager	Soil	Moderate	1.50	1.00	25
Myrmicinae	<i>Crematogaster</i> sp. 2	GenForager	Soil	Moderate	2.00	1.00	12
Myrmicinae	<i>Crematogaster</i> sp. 3	GenForager	Soil	Moderate	1.50	1.00	11
Myrmicinae	<i>Crematogaster</i> sp. 4	GenForager	Soil	Moderate	1.50	1.00	4
Myrmicinae	<i>Crematogaster</i> sp. 5	GenForager	Soil	Moderate	1.50	1.00	49
Myrmicinae	<i>Crematogaster</i> sp. 6	GenForager	Soil	Moderate	1.00	1.00	40
Myrmicinae	<i>Crematogaster</i> sp. 7	GenForager	Soil	Moderate	1.00	0.25	53
Myrmicinae	<i>Crematogaster</i> sp. 8	GenForager	Soil	Moderate	1.00	0.50	34
Myrmicinae	<i>Crematogaster</i> sp. 9	GenForager	Soil	Moderate	1.00	0.50	60
Myrmicinae	<i>Crematogaster</i> sp. 10	GenForager	Soil	Moderate	1.00	0.50	28
Myrmicinae	<i>Crematogaster</i> sp. 11	GenForager	Soil	Moderate	1.00	0.50	30
Myrmicinae	<i>Crematogaster</i> sp. 12	GenForager	Soil	Moderate	1.50	1.00	38
Proceratiinae	<i>Discothyrea</i> sp. 1	Pred	Litter	Small	1.00	0.25	59
Ponerinae	<i>Hypoponera</i> sp. 1	Pred	Litter	Small	2.00	1.00	21
Ponerinae	<i>Leptogenys</i> sp. 1	Pred	Soil	Large	3.00	2.50	2

Abbreviations: GenForager = Generalist forager, Pred = Predator, Sap = Saprovore, Soil = soil-dweller, Litter = litter-dweller, Veg = vegetation-dweller.

Table S2 continued

Subfamily	Genus	Forage strategy	Habitat position	Eye Size	Body length (mm)	Femur length (mm)	Colour value (V)
Ponerinae	<i>Leptogenys</i> sp. 2	Pred	Soil	Large	3.00	2.00	4
Ponerinae	<i>Leptogenys</i> sp. 3	Pred	Soil	Large	2.50	2.00	29
Myrmicinae	<i>Meranoplus</i> sp. 1	GenForager	Soil	Large	1.00	1.00	33
Myrmicinae	<i>Monomorium</i>	GenForager	Litter	Moderate	1.00	0.50	64

	sp. 1						
Myrmicinae	<i>Monomorium</i> sp. 2	GenForager	Litter	Moderate	0.50	0.25	42
Myrmicinae	<i>Monomorium</i> sp. 3	GenForager	Litter	Moderate	0.50	0.25	33
Myrmicinae	<i>Monomorium</i> sp. 4	GenForager	Litter	Moderate	1.00	1.00	71
Myrmicinae	<i>Myrmicaria</i> sp. 1	GenForager	Soil	Large	2.50	3.00	71
Ponerinae	<i>Paltothyreus</i> sp. 1	Pred	Soil	Large	5.00	3.00	8
Ponerinae	<i>Paltothyreus</i> sp. 2	Pred	Soil	Large	5.00	3.00	12
Ponerinae	<i>Parasyscia</i> sp. 1	Pred	Litter	Small	1.00	0.50	52
Ponerinae	<i>Parasyscia</i> sp. 2	Pred	Litter	Small	0.50	0.25	67
Ponerinae	<i>Parasyscia</i> sp. 3	Pred	Litter	Small	1.00	0.50	61
Myrmicinae	<i>Pheidole</i> sp. 1	GenForager	Soil	Small	1.00	1.00	70
Myrmicinae	<i>Pheidole</i> sp. 2	GenForager	Soil	Small	1.50	1.00	77
Myrmicinae	<i>Pheidole</i> sp. 3	GenForager	Soil	Small	1.50	1.00	59
Myrmicinae	<i>Pheidole</i> sp. 4	GenForager	Soil	Small	2.00	1.50	45
Myrmicinae	<i>Pheidole</i> sp. 5	GenForager	Soil	Small	0.50	0.25	36
Myrmicinae	<i>Pheidole</i> sp. 5	GenForager	Soil	Small	1.00	0.50	66
Myrmicinae	<i>Pheidole</i> sp. 6	GenForager	Soil	Small	1.50	1.00	65
Proceratiinae	<i>Probolomyrme</i> x sp. 1	Pred	Litter	Absent	1.00	0.25	59
Proceratiinae	<i>Probolomyrme</i> x sp. 2	Pred	Litter	Absent	1.50	1.00	77
Myrmicinae	<i>Solenopsis</i> sp. 1	GenForager	Soil	Small	1.00	1.00	72
Myrmicinae	<i>Solenopsis</i> sp. 2	GenForager	Soil	Small	1.00	0.25	57
Myrmicinae	<i>Solenopsis</i> sp. 3	GenForager	Soil	Small	1.00	0.50	53
Myrmicinae	<i>Solenopsis</i> sp. 4	GenForager	Soil	Small	0.50	0.25	22
Myrmicinae	<i>Solenopsis</i> sp. 5	GenForager	Soil	Small	1.00	0.25	60
Myrmicinae	<i>Solenopsis</i> sp. 6	GenForager	Soil	Small	0.50	0.30	71
Myrmicinae	<i>Strumigenys</i> sp. 1	Pred	Litter	Moderate	1.00	0.50	6
Myrmicinae	<i>Strumigenys</i> sp. 2	Pred	Litter	Moderate	1.50	0.50	13
Myrmicinae	<i>Strumigenys</i> sp. 3	Pred	Litter	Moderate	2.00	1.00	52
Myrmicinae	<i>Tetramorium</i>	GenForager	Soil	Large	1.00	0.50	6

	sp. 1						
Myrmicinae	<i>Tetramorium</i> sp. 2	GenForager	Soil	Large	0.50	0.50	52
Myrmicinae	<i>Tetramorium</i> sp. 3	GenForager	Soil	Large	1.00	0.50	32
Myrmicinae	<i>Tetramorium</i> sp. 4	GenForager	Soil	Large	0.50	0.50	44
Myrmicinae	<i>Tetramorium</i> sp. 5	GenForager	Soil	Large	3.00	2.50	28
Myrmicinae	<i>Tetramorium</i> sp. 6	GenForager	Soil	Large	1.00	0.50	47
Myrmicinae	<i>Tetramorium</i> sp. 7	GenForager	Soil	Large	1.50	1.00	63
Myrmicinae	<i>Tetramorium</i> sp. 8	GenForager	Soil	Large	1.00	0.50	52
Myrmicinae	<i>Tetramorium</i> sp. 9	GenForager	Soil	Large	1.50	1.00	53
Myrmicinae	<i>Tetramorium</i> sp. 10	GenForager	Soil	Large	2.00	1.00	50
Myrmicinae	<i>Tetramorium</i> sp. 11	GenForager	Soil	Large	1.50	1.50	25
Myrmicinae	<i>Tetramorium</i> sp. 12	GenForager	Soil	Large	2.00	2.00	29
Myrmicinae	<i>Tetramorium</i> sp. 13	GenForager	Soil	Large	1.00	1.00	28
Myrmicinae	<i>Tetramorium</i> sp. 14	GenForager	Soil	Large	1.50	0.50	58
Myrmicinae	<i>Tetramorium</i> sp. 15	GenForager	Soil	Large	1.00	0.50	46
Myrmicinae	<i>Tetramorium</i> sp. 16	GenForager	Soil	Large	1.00	0.50	32
Myrmicinae	<i>Tetramorium</i> sp. 17	GenForager	Soil	Large	1.00	0.50	56
Myrmicinae	<i>Tetramorium</i> sp. 18	GenForager	Soil	Large	1.00	0.50	50
Myrmicinae	<i>Tetramorium</i> sp. 19	GenForager	Soil	Large	1.00	1.00	26
Myrmicinae	<i>Tetramorium</i> sp. 20	GenForager	Soil	Large	1.00	0.50	61
Myrmicinae	<i>Tetramorium</i> sp. 21	GenForager	Soil	Large	1.50	0.50	34
Myrmicinae	<i>Tetramorium</i> sp. 22	GenForager	Soil	Large	1.00	1.00	43
Myrmicinae	<i>Tetramorium</i> sp. 23	GenForager	Soil	Large	1.00	0.50	57
Myrmicinae	<i>Trichomyrmex</i> sp. 1	GenForager	Soil	Moderate	1.00	1.00	49
Myrmicinae	<i>Trichomyrmex</i> sp. 2	GenForager	Soil	Moderate	1.00	0.50	70
Myrmicinae	<i>Trichomyrmex</i> sp. 3	GenForager	Soil	Moderate	1.00	0.50	30

Myrmicinae	<i>Trichomyrmex</i> sp. 4	GenForager	Soil	Moderate	1.00	1.00	68
Myrmicinae	<i>Trichomyrmex</i> sp. 5	GenForager	Soil	Moderate	1.00	1.50	49
Myrmicinae	<i>Trichomyrmex</i> sp. 6	GenForager	Soil	Moderate	1.50	1.00	66

Abbreviations: GenForager = Generalist forager, Pred = Predator, Sap = Saprovore, Soil = soil-dweller, Litter = litter-dweller, Veg = vegetation-dweller.

Table S3 References used to establish genus-level trait identities of ants in the Midlands and Zululand, as well as for the justification of traits used in this study. References are arranged alphabetically.

Author(s)	Year	Title	Journal/Book
Alpert	1992	Observations on the genus <i>Terataner</i> in Madagascar (Hymenoptera: Formicidae)	Psyche 99: 117-127
Andersen	1988	Immediate and longer-term effects of fire on seed predation by ants in sclerophyllous vegetation in south-eastern Australia	Australian Journal of Ecology 13: 285-293
Andersen	1995	A classification of Australian ant communities, based on functional groups which parallel plant life-forms in relation to stress and disturbance	Journal of Biogeography 22: 15-29
Andersen	2006	A systematic overview of Australian species of the myrmicine ant genus <i>Meranoplus</i> F. Smith, 1853 (Hymenoptera: Formicidae)	Myrmecologische Nachrichten 8: 157-170
Arcusa	2019	Fire effects on the ant community in areas of native and exotic vegetation	Sociobiology 66: 44-51
Arenas-Clavijo and Armbrrecht	2018	Guilds and diversity of ants in three land uses from a coffee landscape at Cauca-Colombia	Revista de Biología Tropical 66: 48-57
Arnan et al.	2013	Response of ant functional composition to fire	Ecography 36: 1182-1193
Berg et al.	1998	Dynamics and stratification of functional groups of micro-and mesoarthropods in the organic layer of a Scots pine forest	Biology and Fertility of Soils 26: 268-284
Bihn et al.	2010	Loss of functional diversity of ant assemblages in secondary tropical forests	Ecology 91: 782-792
Bihn et al.	2008	Do secondary forests act as refuges for old growth forest animals? Recovery of ant diversity in the Atlantic forest of Brazil	Biological Conservation 141: 733-743
Bishop et al.	2016	Ant assemblages have darker and larger members in cold environments	Global Ecology and Biogeography 25: 1489-1499
Bishop et al.	2015	Contrasting species and functional beta diversity in montane ant assemblages	Journal of Biogeography 42: 1776-1786
Blanchard and Moreau	2017	Defensive traits exhibit an evolutionary trade-off and drive diversification in ants	Evolution 71: 315-328
Boet et al.	2020	The role of environmental vs. biotic filtering in the structure of European ant communities: A matter of trait type and spatial scale	PloS One 15: e0228625
Boi et al.	2000	Modeling the aggregative behavior of ants of the species <i>Polyergus rufescens</i>	Nonlinear Analysis: Real World Applications 1: 163-176
Bolton and Marsh	1989	The Afrotropical thermophilic ant genus <i>Ocymyrmex</i> (Hymenoptera: Formicidae)	Journal of Natural History 23: 1267-1308
Bolton and Fisher	2008	Afrotropical ants of the ponerine genera <i>Centromyrmex</i> Mayr, <i>Promyopias</i> Santschi gen. rev. and <i>Feroponera</i> gen. n., with a revised key to genera of African Ponerinae (Hymenoptera: Formicidae)	Zootaxa 1929: 1-37
Bolton and Fisher	2011	Taxonomy of Afrotropical and West Palaearctic ants of the ponerine genus <i>Hypoponera</i> Santschi (Hymenoptera: Formicidae)	Zootaxa 2843: 1-118
Bolton	1973	The ant genera of West Africa: A synonymic synopsis with keys (Hymenoptera: Formicidae)	Bulletin of the British Museum (Natural History)

			Entomology 27: 317-368
Bolton	1974	A revision of the ponerine ant genus <i>Plectroctena</i> F. Smith (Hymenoptera: Formicidae)	Bulletin of the British Museum (Natural History) Entomology 30: 309-338
Bolton	1981	A revision of six minor genera of Myrmicinae (Hymenoptera: Formicidae) in the Ethiopian zoogeographical region	Bulletin of the British Museum (Natural History) Entomology 43: 245-307
Bolton	1981	A revision of the ant genera <i>Meranoplus</i> F. Smith, <i>Dicroaspis</i> Emery and <i>Calyptomyrmex</i> Emery (Hymenoptera: Formicidae) in the Ethiopian zoogeographic region	Bulletin of the British Museum (Natural History) Entomology 42: 43-81
Bolton	1987	A review of the <i>Solenopsis</i> genus-group and revision of Afrotropical <i>Monomorium</i> Mayr (Hymenoptera: Formicidae)	Bulletin of the British Museum (Natural History). Entomology 54: 263-452
Borowiec	2016	Generic revision of the ant subfamily Dorylinae (Hymenoptera, Formicidae)	ZooKeys 608: 1-120
Branstetter et al.	2017	Dry habitats were crucibles of domestication in the evolution of agriculture in ants	Proceedings of the Royal Society B: Biological Sciences 284: 20170095

Table S3 continued

Author(s)	Year	Title	Journal/Book
Brown and Wilson	1959	The evolution of the dacetine ants	The Quarterly Review of Biology 34: 278-294
Brown	1958	Contributions toward a reclassification of the Formicidae. II. Tribe Ectatommini (Hymenoptera)	Bulletin of the Museum of Comparative Zoology 118: 173-362
Brown	1976	Contributions toward a reclassification of the Formicidae. Part VI. Ponerinae, tribe Ponerini, subtribe Odontomachiti. Section A. Introduction, subtribal characters. Genus <i>Odontomachus</i>	Studia Entomologica 19: 67-171
Brown	1978	Contributions toward a reclassification of the Formicidae. Part VI. Ponerinae, tribe Ponerini, subtribe Odontomachiti. Section B. Genus <i>Anochetus</i> and bibliography	Studia Entomologica 20: 549-638
Brown	1979	A remarkable new species of <i>Proceratium</i> , with dietary and other notes on the genus (Hymenoptera: Formicidae)	Psyche 86: 337-346
Carvalho et al.	2020	Understanding what bioindicators are actually indicating: Linking disturbance responses to ecological traits of dung beetles and ants	Ecological Indicators 108: 105764
Castro-Souza et al.	2019	Drivers of ant composition, richness, and trophic guilds in Neotropical iron ore cavities	International Journal of Speleology 48: 7
Cover and Deyrup	2007	A new ant genus from the southwestern United States	Memoirs of the American Entomological Institute 80: 89-99
de Carvalho Moretti et al.	2011	<i>Pycnoscelus surinamensis</i> (Linnaeus, 1758) (Blaberoidea: Blaberidae), a cockroach with a possible association with the ant <i>Brachymyrmex cordemoyi</i> Forel, 1895 (Hymenoptera: Formicidae) and which may be exhibiting a domiciliation trend	Symbiosis 53: 37-39

de Souza-Campana et al.	2016	Epigaeic ant (Hymenoptera: Formicidae) communities in urban parks located in Atlantic Forest biome	Check List 12: 1967
Dejean et al.	2012	The ecology and feeding habits of the arboreal trap-jawed ant <i>Daceton armigerum</i>	PLoS One 7: e37683
Delabie et al.	2000	Litter ant communities of the Brazilian Atlantic rain forest region	Curtin University of Technology School of Environmental Biology Bulletin 18: 1-17
DiGirolamo and Fox	2006	The influence of abiotic factors and temporal variation on local invasion patterns of the Argentine ant (<i>Linepithema humile</i>)	Biological Invasions 8: 125-135
Diniz and Brandão	1993	Biology and myriapod egg predation by the Neotropical myrmicine ant <i>Stegomyrmex vizottoi</i> (Hymenoptera: Formicidae)	Insectes Sociaux, 40: 301-311
Donoso	2012	Additions to the taxonomy of the armadillo ants (Hymenoptera, Formicidae, Tatuidris)	Zootaxa 3503: 61-81
Dubovikoff and Longino	2004	A new species of the genus <i>Bothriomyrmex</i> Emery, 1869 (Hymenoptera: Formicidae: Dolichoderinae) from Costa Rica	Zootaxa 776: 1-10
Elmes	1991	The social biology of <i>Myrmica</i> ants	Actes des Colloques Insectes Sociaux 7: 17-34
Feener et al.	1988	Curvilinear allometry, energetics and foraging ecology: A comparison of leafcutting ants and army ants	Functional Ecology 2: 509-520
Fernández	2003	Revision of the myrmicine ants of the <i>Adelomyrmex</i> genus-group (Hymenoptera: Formicidae)	Zootaxa 361: 1-52
Fiedler et al.	2007	Stable N-isotope signatures of central European ants—assessing positions in a trophic gradient	Insectes Sociaux, 54: 393-402
Fischer et al.	2014	The ant genus <i>Carebara</i> Westwood (Hymenoptera, Formicidae): Synonymisation of <i>Pheidologeton</i> Mayr under <i>Carebara</i> , establishment and revision of the <i>C. polita</i> species group	ZooKeys 438: 57-112
Fisher and Robertson	1999	Silk production by adult workers of the ant <i>Melissotarsus emeryi</i> (Hymenoptera, Formicidae) in South African fynbos	Insectes Sociaux 46: 78-83
Fisher and Smith	2008	A revision of Malagasy species of <i>Anochetus</i> Mayr and <i>Odontomachus</i> Latreille (Hymenoptera: Formicidae)	PLoS ONE 3: 1-23
Fisher and Bolton	2016	Ants of Africa and Madagascar: A guide to the genera	University of California Press, Oakland, California
Fitzpatrick et al.	2011	Forecasting the future of biodiversity: A test of single-and multi-species models for ants in North America	Ecography 34: 836-847
Fjellberg	1985	Recent advances and future needs in the study of <i>Collembola</i> biology and systematics	Quaestiones Entomologicae 21: 559-570
Garcia et al.	2013	The ants of Kenya (Hymenoptera: Formicidae: Faunal overview, first species checklist, bibliography, accounts for all genera, and discussion on taxonomy and zoogeography	Journal of East African Natural History 101: 127-222

Table S3 continued

Author(s)	Year	Title	Journal/Book
Gronenberg	1996	The trap-jaw mechanism in the dacetine ants <i>Daceton armigerum</i> and <i>Strumigenys</i> sp.	Journal of Experimental Biology 199: 2021-2033
Guilherme et al.	2019	Can environmental complexity predict functional trait composition of ground-dwelling ant assemblages? A test across	Acta Oecologica 99: 103434

Heinze et al.	2006	Stealthy invaders: The biology of Cardiocondyla tramp ants	Insectes Sociaux 53: 1-7
Helms et al.	2020	Ant biodiversity and ecosystem services in bioenergy landscapes	Agriculture, Ecosystems and Environment 290: 106780
Hernández et al.	2010	Checklist, biological notes and distribution of ants (hymenoptera: formicidae) from barranca de metztitlan biosphere reserve, hidalgo, mexico	Sociobiology 56: 397-434
Hlongwane	2018	Use of ground-dwelling arthropods as bioindicators of ecological condition in grassland and forest vegetation at eThekweni Municipality in KwaZulu-Natal	Doctoral dissertation
Hölldobler and Lumsden	1980	Territorial strategies in ants	Science 210: 732-739
Hölldobler and Wilson	1990	The Ants	Harvard University Press, Cambridge
Hölldobler	1981	Foraging and spatiotemporal territories in the honey ant <i>Myrmecocystus mimicus</i> Wheeler (Hymenoptera: Formicidae)	Behavioural Ecology and Sociobiology 9: 301-314
Hopkin	1997	Biology of the Springtails (Insecta: Collembola)	Oxford University Press, Oxford
Hyodo et al.	2011	Feeding habits of Hymenoptera and Isoptera in a tropical rain forest as revealed by nitrogen and carbon isotope ratios	Insectes Sociaux 58: 417
Ito	2001	Specialized predation on arthropod eggs in a myrmicine ant, <i>Calyptomyrmex</i> sp. (Hymenoptera: Formicidae), in the oriental tropics	Tropics 10: 405-407
Ito	1998	Colony composition and specialized predation on millipedes in the enigmatic ponerine ant genus <i>Probolomyrmex</i> (Hymenoptera, Formicidae)	Insectes Sociaux 45: 79-83
Kaspari and Weiser	1999	The size-grain hypothesis and interspecific scaling in ants	Functional Ecology 13: 530-538
Katayama	2013	Predatory behaviours of <i>Discothyrea kamiteta</i> (Proceratiinae) on spider eggs	Asian Myrmecol 5: 121-124
Kenne et al.	2009	Nesting and foraging habits in the arboreal ant <i>Atopomyrmex mocquersyi</i> André (Formicidae, Myrmicinae)	Myrmecological News 12: 109-115
Klingenberg and Brandao	2009	Revision of the fungus-growing ant genera <i>Mycetophylax</i> Emery and <i>Paramycetophylax</i> Kusnezov rev. stat., and description of <i>Kalathomyrmex</i> n. gen. (Formicidae: Myrmicinae: Attini)	Zootaxa 2052: 1-31
Koch et al.	2019	Comparative evaluation of taxonomic and functional diversities of leaf-litter ants of the Brazilian Atlantic Forest	Turkish Journal of Zoology 43: 437-546
Koch and Vohland	2004	Ants along a southern African transect-a basis for biodiversity change monitoring (Insecta, Hymenoptera, Formicidae)	Zoosystematics and Evolution 80: 261-273
Lach et al.	2010	Ant ecology	Oxford University Press, New York
LaPolla et al.	2011	Monograph of <i>Nylanderia</i> (Hymenoptera: Formicidae) of the World: An introduction to the systematics and biology of the genus	Zootaxa 3110: 1-9
LaPolla et al.	2010	Phylogeny and taxonomy of the <i>Prenolepis</i> genus-group of ants (Hymenoptera: Formicidae)	Systematic Entomology 35: 118-131
LaPolla et al.	2002	Natural history of the mealybug-tending ant <i>Acropyga epedana</i> , with descriptions of the male and queen castes (Hymenoptera:	Transactions of the American Entomological Society 128:

		Formicidae)	367-376
Leal et al.	2012	Effects of habitat fragmentation on ant richness and functional composition in Brazilian Atlantic forest	Biodiversity and Conservation 21: 1687-1701
Levieux	1972	Le rôle des fourmis dans les réseaux trophiques d'une savane préforestière de Côte d'Ivoire	Annales de l'Université d'Abidjan 5: 143-240
Levieux	1983	The soil fauna of tropical savannas	Elsevier, Amsterdam: 525-540
Liefke et al.	1998	Nesting and food resources of syntopic species of the ant genus <i>Polyrhachis</i> (Hymenoptera, formicidae) in West-Malaysia	Insectes Sociaux 45: 411-425
Longino	2003	The <i>Crematogaster</i> (Hymenoptera, Formicidae, Myrmicinae) of Costa Rica	Zootaxa 151: 1-150

Table S3 continued

Author(s)	Year	Title	Journal/Book
Lyu et al.	2002	Review of Korean Dacetini (Hymenoptera, Formicidae, Myrmicinae)	Insecta Koreana 18: 229-241
Majer and Delabie	1994	Comparison of the ant communities of annually inundated and terra firme forests at Trombetas in the Brazilian Amazon	Insectes Sociaux 41: 343-359
Marques and Del-Claro	2006	The ant fauna in a Cerrado area: The influence of vegetation structure and seasonality (Hymenoptera: Formicidae)	Sociobiology 47: 235-252
Masuko	1990	Behavior and ecology of the enigmatic ant <i>Leptanilla japonica</i> Baroni Urbani (Hymenoptera: Formicidae: Leptanillinae)	Insectes Sociaux 37: 31-57
Micolino et al.	2019	Chromosomal dynamics in space and time: Evolutionary history of <i>Mycetophylax</i> ants across past climatic changes in the Brazilian Atlantic coast	Scientific Reports 9: 1-13
Montine et al.	2014	Seasonality of epigaeic ant communities in a Brazilian Atlantic Rainforest	Sociobiology 61: 178-183
Moretti et al.	2017	Handbook of protocols for standardized measurement of terrestrial invertebrate functional traits	Functional Ecology 31: 558-567
Ness et al.	2004	Ant body size predicts dispersal distance of ant-adapted seeds: Implications of small-ant invasions	Ecology 85: 1244-1250
Pacheco et al.	2009	A comparison of the leaf-litter ant fauna in a secondary atlantic forest with an adjacent pine plantation in southeastern Brazil	Neotropical Entomology 38: 55-65
Paul and Roces	2003	Fluid intake rates in ants correlate with their feeding habits	Journal of Insect Physiology 49: 347-357
Pekár and Mayntz	2014	Comparative analysis of the macronutrient content of Central European ants (Formicidae): Implications for ant-eating predators	Journal of insect physiology 62: 32-38
Perkovsky	2016	Rovno Amber ant assemblage: Bias toward arboreal strata or sampling effect?	Vestnik Zoologii 50: 271-274
Ponge et al.	2006	Decreased biodiversity in soil springtail communities: The importance of dispersal and landuse history in heterogeneous landscapes	Soil Biology and Biochemistry 38: 1158-1161
Pratt	1994	Ecology and behavior of <i>Gnamptogenys horni</i> (Formicidae:	Insectes Sociaux 41: 255-262

Ponerinae)

Prins	1982	Review of Anoplolepis with reference to male genitalia, and notes on Acropyga (Hymenoptera, Formicidae)	Annals of the South African Museum 89: 235-247
Probst et al.	2019	A phylogenetic analysis of the dirt ants, Basiceros (Formicidae: Myrmicinae): Inferring life histories through morphological convergence	Insect Systematics and Diversity 3: 3
Rabeling et al.	2016	<i>Lenomyrmex hoelldobleri</i> : A new ant species discovered in the stomach of the dendrobatid poison frog, <i>Oophaga sylvatica</i> (Funkhouser)	ZooKeys 618: 79
Radchenko and Elmes	1998	Taxonomic revision of the ritae species-group of the genus <i>Myrmica</i> (Hymenoptera, Formicidae)	Vestnik Zoologii 32: 3-27
Raymond-Léonard et al.	2019	A novel set of traits to describe Collembola mouthparts: Taking a bite out of the broad chewing mandible classification	Soil Biology and Biochemistry 138: 107608
Ribeiro et al.	2019	Different trophic groups of arboreal ants show differential responses to resource supplementation in a neotropical savanna	Oecologia 190: 433-443
Ríos-Casanova et al.	2006	Ant diversity and its relationship with vegetation and soil factors in an alluvial fan of the Tehuacán Valley, Mexico	Acta Oecologica 29: 316-323
Rivas-Arancibia et al.	2014	Effect of disturbance on the ant community in a semiarid region of central México	Applied Ecology and Environmental Research 12: 703-716
Robson and Kohout	2007	A review of nesting habits and socioecology of the ant genus <i>Polyrhachis</i> Smith	Asian Myrmecology 1: 81-99
Rosado et al.	2012	Epigeic ants (Hymenoptera: Formicidae) in vineyards and grassland areas in the Campanha region, state of Rio Grande do Sul, Brazil	Check List 8: 1184
Ross	1994	The phytophagous insect community on the veld fig, <i>Ficus burtt-davyi</i> Hutch	Masters dissertation
Ruiz et al.	2017	Genetic variation in light vision and light-dependent movement behaviour in the eyeless Collembola <i>Folsomia candida</i>	Pedobiologia 61: 33-41
Salmon et al.	2014	Linking species, traits and habitat characteristics of Collembola at European scale	Soil Biology and Biochemistry 75: 73-85
Santschi	1906	A propos de mœurs parasitiques temporaires des fourmis du genre <i>Bothriomyrmex</i>	Annales de la Société Entomologique de France 75: 363-392 Zootaxa 3817: 1-242

Table S3 continued

Author(s)	Year	Title	Journal/Book
Schmidt and Shattuck	2014	The higher classification of the ant subfamily Ponerinae (Hymenoptera: Formicidae), with a review of ponerine ecology and behavior	
Schofield et al.	2016	Morphological characteristics of ant assemblages (Hymenoptera: Formicidae) differ among contrasting biomes	Myrmecological News 23: 129-137
Seifert	2007	Die Ameisen Mittel- und Nordeuropas	Lutra, Klitten: 368
Shattuck	1991	Revision of the dolichoderine ant genus <i>Axinidris</i> Hymenoptera:	Systematic Entomology 16:

		Formicidae	105-120
Shattuck	2011	Revision of the ant genus <i>Calyptomyrmex</i> (Hymenoptera: Formicidae) in South-east Asia and Oceania	Zootaxa 2743: 1-26
Slingsby	2017	Ants of Southern Africa: The ant book for all	Slingsby Maps, Muizenberg
Snelling	2007	A review of the arboreal Afrotropical ant genus <i>Axinidris</i>	Memoirs of the American Entomological Institute 80: 551-579
Staab et al.	2014	Tree diversity promotes predator but not omnivore ants in a subtropical Chinese forest	Ecological Entomology 39: 637-647
Susilo et al.	2019	Diversity and abundance of ant (Formicidae) functional groups in range of land use systems in Jambi, Sumatra	Biota: Jurnal Ilmiah Ilmu-Ilmu Hayati 15: 8-16
Taylor	1965	The Australian ants of the genus <i>Pristomyrmex</i> , with a case of apparent character displacement	Psyche 72: 35-54
Vicente et al.	2011	New distribution record of <i>Daceton boltoni</i> Azorsa and Sosa-Calvo, 2008 (Insecta: Hymenoptera) in the Brazilian Amazon	Check List 7: 878-879
Wang	2003	A monographic revision of the ant genus <i>Pristomyrmex</i> (Hymenoptera: Formicidae)	Bulletin of the Museum of Comparative Zoology 157: 383-539
Ward	1991	Phylogenetic analysis of pseudomyrmecine ants associated with domatiabearing plants	Oxford University Press, Oxford: 335-352
Ward	2001	Taxonomy, phylogeny and biogeography of the ant genus <i>Tetraponera</i> (Hymenoptera: Formicidae) in the Oriental and Australian regions	Invertebrate Systematics 15: 589-665
Ward et al.	2015	Zur Taxonomie der Gattung <i>Chalepoxenus</i> (Hymenoptera, Formicidae, Myrmicinae)	Systematic Entomology 40: 129-148
Weber	1949	The functional significance of dimorphism in the African ant, <i>Oecophylla</i>	Ecology 30: 397-400
Weiser and Kaspari	2006	Ecological morphospace of New World ants	Ecological Entomology 31: 131-142
Wetterer	2020	First North American records of the old-world tramp ant <i>Sylophopsis sechellensis</i> (Hymenoptera: Formicidae)	Sociobiology 67: 473-477
Wetterer and Sharaf	2017	Sharaf M.R. (2017). Worldwide distribution of <i>Sylophopsis sechellensis</i> (Hymenoptera: Formicidae)	Florida Entomologist 100: 281-285
Wheeler	1910	Ants: Their structure, development and behavior	Columbia University Press New York: 663
Wheeler	1922	Ants collected by the American Museum Congo Expedition. A contribution to the myrmecology of Africa	Bulletin of the American Museum of Natural History 45: 1-1055
Whitford et al.	2008	Effects of three species of Chihuahuan Desert ants on annual plants and soil properties	Journal of Arid Environments 72: 392-400
Wilkie et al.	2007	Biodiversity below ground: Probing the subterranean ant fauna of Amazonia	Naturwissenschaften 94: 725-731
Wilson and Brown	1984	Behavior of the cryptobiotic predaceous ant <i>Eurhopalothrix heliscata</i> , n. sp. (Hymenoptera: Formicidae: Basicerotini)	Insectes Sociaux 31: 408-428
Zimmerli and Topoff	1994	Queens of the socially parasitic ant <i>Polyergus</i> do not kill queens of <i>Formica</i> that have not formed colonies (Hymenoptera: Formicidae)	Journal of Insect Behavior 7: 119-121

Table S4 Species list and trait identities used in the analysis of springtails in the Midlands and Zululand.

Suborder	Family	Furca development	Body length (mm)	Eye size	Habitat position	Feeding strategy
Entomobryomorpha	Entomobryidae sp.1	Long	1.35	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.2	Long	2.15	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.3	Long	2.00	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.4	Long	1.50	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.5	Long	1.50	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.6	Long	1.00	Moderate	Veg	Chew
Entomobryomorpha	Entomobryidae sp.7	Long	1.38	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.8	Long	1.00	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.9	Long	0.86	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.10	Long	0.80	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.11	Long	0.90	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.12	Long	1.00	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.13	Long	1.50	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.14	Long	1.88	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.15	Long	1.05	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.16	Long	1.38	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.17	Long	1.50	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.18	Long	0.75	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.19	Long	0.80	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.20	Long	1.25	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.21	Long	1.30	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.22	Long	1.35	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.23	Long	1.38	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.24	Long	0.92	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.25	Long	0.63	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.26	Long	1.38	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.27	Long	0.46	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.28	Long	1.13	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.29	Long	0.53	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.30	Long	1.25	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.31	Long	1.30	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.32	Long	0.93	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.33	Long	1.08	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.34	Long	1.33	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.35	Long	1.40	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.36	Long	1.08	Large	Veg	Chew

Entomobryomorpha	Entomobryidae sp.37	Long	0.40	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.38	Long	0.80	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.39	Long	0.24	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.40	Long	1.38	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.41	Long	0.80	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.42	Long	0.28	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.43	Long	0.80	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.44	Long	1.05	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.45	Long	1.13	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.46	Long	1.15	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.47	Long	0.93	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.48	Long	0.93	Small	Litter	Chew
Entomobryomorpha	Entomobryidae sp.49	Long	0.60	Large	Litter	Chew
Entomobryomorpha	Entomobryidae sp.50	Long	1.30	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.51	Long	1.13	Large	Veg	Chew
Entomobryomorpha	Entomobryidae sp.52	Long	0.93	Large	Veg	Chew
Entomobryomorpha	Isotomidae sp.1	Moderate	1.10	Moderate	Litter	Chew
Entomobryomorpha	Isotomidae sp.2	Moderate	0.70	Large	Litter	Chew
Entomobryomorpha	Isotomidae sp.3	Moderate	0.90	Moderate	Litter	Chew
Entomobryomorpha	Isotomidae sp.4	Moderate	0.75	Large	Litter	Chew
Entomobryomorpha	Isotomidae sp.5	Moderate	0.43	Large	Litter	Chew
Poduromorpha	Brachystomellidae sp.1	Short	0.56	Large	Litter	Suck
Poduromorpha	Hypogastruridae sp.1	Short	0.80	Large	Litter	Chew
Poduromorpha	Hypogastruridae sp.2	Short	0.30	Large	Litter	Chew
Poduromorpha	Hypogastruridae sp.3	Short	0.58	Large	Litter	Chew
Poduromorpha	Hypogastruridae sp.4	Short	0.60	Large	Litter	Chew
Poduromorpha	Hypogastruridae sp.5	Short	1.00	Large	Litter	Chew
Poduromorpha	Hypogastruridae sp.6	Short	1.00	Large	Litter	Chew
Poduromorpha	Hypogastruridae sp.7	Short	1.04	Large	Litter	Chew
Poduromorpha	Neanuridae sp.1	Short	0.54	Large	Litter	Suck
Poduromorpha	Neanuridae sp.2	Short	0.48	Absent	Soil	Suck
Poduromorpha	Neanuridae sp.3	Short	0.36	Large	Litter	Suck
Poduromorpha	Neanuridae sp.4	Short	0.60	Large	Litter	Suck
Poduromorpha	Neanuridae sp.5	Short	0.92	Large	Litter	Suck
Poduromorpha	Neanuridae sp.6	Short	0.56	Large	Litter	Suck

Abbreviations: Soil = soil-dweller, Litter = litter-dweller, Veg = vegetation-dweller, Chew = chewing mandibles, Suck = sucking mandibles

Table S4 continued

Suborder	Family	Furca development	Body length (mm)	Eye size	Habitat position	Feeding strategy
Poduromorpha	Odontellidae sp.1	Short	0.36	Large	Litter	Suck
Poduromorpha	Onychiuridae sp.1	Absent	1.84	Absent	Soil	Chew
Symphyleona	Dicyrtomidae sp.1	Long	1.20	Large	Veg	Chew
Symphyleona	Dicyrtomidae sp.2	Long	0.88	Large	Veg	Chew
Symphyleona	Dicyrtomidae sp.3	Long	0.40	Large	Veg	Chew
Symphyleona	Dicyrtomidae sp.4	Long	0.26	Large	Veg	Chew
Symphyleona	Dicyrtomidae sp.5	Long	0.46	Large	Veg	Chew
Symphyleona	Dicyrtomidae sp.6	Long	1.10	Large	Veg	Chew
Symphyleona	Dicyrtomidae sp.7	Long	0.66	Large	Veg	Chew
Symphyleona	Katiannidae sp.1	Long	0.38	Large	Veg	Chew
Symphyleona	Katiannidae sp.2	Long	0.32	Large	Veg	Chew
Symphyleona	Katiannidae sp.3	Long	1.12	Large	Veg	Chew
Symphyleona	Katiannidae sp.4	Long	1.00	Large	Veg	Chew
Symphyleona	Katiannidae sp.5	Long	1.00	Large	Veg	Chew
Symphyleona	Katiannidae sp.6	Long	0.50	Large	Veg	Chew
Symphyleona	Katiannidae sp.7	Long	0.80	Large	Veg	Chew
Symphyleona	Katiannidae sp.8	Long	0.80	Large	Veg	Chew
Symphyleona	Katiannidae sp.9	Long	0.38	Large	Veg	Chew
Symphyleona	Katiannidae sp.10	Long	0.50	Large	Veg	Chew
Symphyleona	Katiannidae sp.11	Long	0.28	Large	Veg	Chew
Symphyleona	Katiannidae sp.12	Long	0.60	Large	Veg	Chew
Symphyleona	Sminthuridae sp.1	Long	1.28	Large	Veg	Chew
Symphyleona	Sminthuridae sp.2	Long	0.66	Large	Veg	Chew
Symphyleona	Sminthuridae sp.3	Long	0.60	Large	Veg	Chew
Symphyleona	Sminthuridae sp.4	Long	0.50	Large	Veg	Chew
Symphyleona	Sminthuridae sp.1	Long	0.92	Large	Veg	Chew
Symphyleona	Sminthuridae sp.2	Long	0.90	Large	Veg	Chew
Symphyleona	Sminthuridae sp.3	Long	0.98	Large	Veg	Chew

Abbreviations: Soil = soil-dweller, Litter = litter-dweller, Veg = vegetation-dweller, Chew = chewing mandibles, Suck = sucking mandibles

Table S5 Justification for springtail (Collembola) and ant (Formicidae) traits included in the analysis. Traits selected have either ecological or physiological significance.

Taxon	Trait	Description	Measurement	Reference
<i>Collembola</i>	Body size	Correlated with dispersal ability, habitat preference and drought tolerance of the individual.	Continuous (mm)	Berg et al. (1998) and Ponge et al. (2006)
	Furca development	The presence and size of the furca is correlated with the individual's dispersal ability and habitat preference.	Categorical (Absent/Short and Moderate/Long)	Ponge et al. (2006) and Salmon et al. (2014)
	Eye size	Play a sensory role and is correlated with light sensitivity, habitat preference and defence against predators.	Categorical (Absent/Small and Moderate/Large)	Hopkin (1997), Salmon et al. (2014) and Ruiz et al. (2017)
	Feeding strategy	Correlated with food preference. Type and size of mouthpart can restrict the type and size of resource that can be ingested.	Categorical (Chewing, Sucking)	Fjellberg (1985)
	Habitat position	Correlated with habitat preference, ecophysiology, and dispersal ability.	Categorical (soil-, litter- or vegetation-dwelling)	Ponge et al. (2006)
<i>Formicidae</i>	Body size	Correlated with the dispersal ability, amount and type resources exploited and physiological/ecological tolerances, which is associated with habitat complexity.	Continuous (mm)	Kaspari and Weiser (1999), Ness et al. (2004), Bihn et al. (2010), Arnan et al. (2013) and Bishop et al. (2015)
	Colour	Correlated with habitat preference and climate tolerance.	Continuous (V)	Bishop et al. (2016)
	Femur length	Correlated with dispersal ability, foraging efficiency and coping with foraging surface temperature, which is associated with habitat complexity.	Continuous (mm)	Feener et al. (1988), Kaspari and Weiser (1999), Weiser and Kaspari (2006), Bihn et al. (2010) and Guilherme et al. (2019).
	Eye size	Correlated with dispersal ability, feeding behaviour (when and where) and efficiency.	Categorical (Absent/Small and Moderate/Large)	Weiser and Kaspari (2006) and Moretti et al. (2017)
	Feeding strategy	Correlated with type and size of food resource exploited.	Categorical (Generalist forager, Predator, Saprovore)	Boet et al. (2020)
	Habitat position	Correlated with habitat/microclimatic preference and feeding behaviour.	Categorical (soil-, litter- or vegetation-dwelling)	Bihn et al. (2008)

See Table S3 for full references.

Table S6 Mean with standard error (SE) and deviation (SD) of all recorded environmental variables within the indigenous forests, grasslands, pine and eucalypt compartments in the Midlands and Zululand. Means with letters in common between biotope types (per region), are not significantly different at $P < 0.05$.

Midlands	Indigenous forest			Grassland			Eucalypt compartment		
	Mean	SE	SD	Mean	SE	SD	Mean	SE	SD
Bare ground cover (%)	0.00 a	0.00	0.00	8.89 b	1.59	6.76	1.11 c	0.65	2.74
Dead wood cover (%)	14.72 a	1.54	6.52	0.56 b	0.56	2.36	56.11 c	2.93	12.43
Grass cover (%)	18.61 a	4.66	19.76	68.61 b	3.59	15.22	0.83 c	0.61	2.57
Herbaceous cover (%)	46.39 a	5.33	22.61	18.33 b	3.26	13.83	1.39 c	0.68	2.87
Leaf litter cover (%)	30.83 a	5.18	21.98	3.06 b	0.59	2.51	96.11 c	1.03	4.39
Leaf litter depth (cm)	4.83 a	0.12	0.51	0.94 b	0.21	0.87	3.00 c	0.00	0.00
Number of plant species	7.72 a	0.55	2.32	5.78 b	0.30	1.26	0.56 c	0.22	0.92
Rock cover (%)	9.72 a	1.83	7.76	7.50 a	2.60	11.01	0.83 b	0.45	1.92
Shade cover (%)	70.56 a	3.21	13.60	0.00 b	0.00	0.00	36.11 c	1.83	7.78
Shrub cover (%)	7.78 a	1.41	6.00	0.56 b	0.38	1.62	0.00 b	0.00	0.00
Soil compaction (psi)	126.33 a	8.85	37.53	65.33 b	4.15	17.63	81.11 b	5.69	24.13
Soil moisture (%)	26.39 a	2.99	12.70	30.56 a	3.08	13.05	35.56 a	3.81	16.17
Soil pH	6.09 a	0.14	0.59	5.81 a	0.04	0.19	5.32 b	0.14	0.61
Tree density (%)	52.50 a	4.21	17.84	0.00 b	0.00	0.00	26.11 c	1.18	5.02
Vegetation cover (%)	69.17 a	5.18	21.98	88.06 b	1.62	6.89	2.78 c	1.01	4.28
Vegetation height (cm)	45.00 a	2.49	10.57	39.44 a	3.02	12.82	8.33 b	3.05	12.95

Zululand	Indigenous forest			Grassland			Eucalypt compartment		
	Mean	SE	SD	Mean	SE	SD	Mean	SE	SD
Bare ground cover (%)	0.28 a	0.28	1.18	10.83 b	2.22	9.43	0.56 a	0.38	1.62
Dead wood cover (%)	25.56 a	2.71	11.49	0.00 b	0.00	0.00	26.11 a	3.15	13.35
Grass cover (%)	10.28 a	1.31	5.55	78.33 b	3.23	13.72	8.89 a	1.74	7.39
Herbaceous cover (%)	44.44 a	4.22	17.90	4.72 b	1.31	5.55	16.67 b	4.78	20.29
Leaf litter cover (%)	40.83 a	5.21	22.11	5.83 b	1.09	4.62	70.00 c	6.35	26.95
Leaf litter depth (cm)	3.06 a	0.24	1.00	0.94 b	0.15	0.64	3.17 a	0.22	0.92
Number of plant species	5.56 a	0.37	1.58	4.06 b	0.29	1.21	2.17 c	0.36	1.54
Rock cover (%)	0.00 n/a	0.00	0.00	0.00 n/a	0.00	0.00	0.00 n/a	0.00	0.00
Shade cover (%)	56.67 a	2.29	9.70	0.00 b	0.00	0.00	44.44 c	1.66	7.05
Shrub cover (%)	8.61 a	1.80	7.63	0.00 b	0.00	0.00	6.67 ab	3.31	14.04
Soil compaction (psi)	94.67 a	2.55	10.80	84.67 a	4.17	17.69	158.00 b	10.86	46.06
Soil moisture (%)	47.78 a	3.92	16.65	27.22 b	2.89	12.27	26.11 b	3.15	13.35
Soil pH	5.46 a	0.15	0.62	5.86 a	0.18	0.77	5.71 a	0.13	0.54
Tree density (%)	44.44 a	3.98	16.88	0.00 b	0.00	0.00	33.33 c	1.14	4.85
Vegetation cover (%)	58.89 a	5.11	21.66	83.33 b	2.89	12.25	29.44 c	6.44	27.33
Vegetation height (cm)	50.28 a	7.66	32.52	17.78 b	2.66	11.27	11.67 b	1.94	8.22

*n/a: Not applicable

Figure S1 Species accumulation curves of all species collected across all biotopes (indigenous forest, grassland, pine and eucalypt compartments) in the Midlands and Zululand. Illustrated is a) Collembola and Formicidae in the Midlands, and b) Collembola and Formicidae in Zululand.

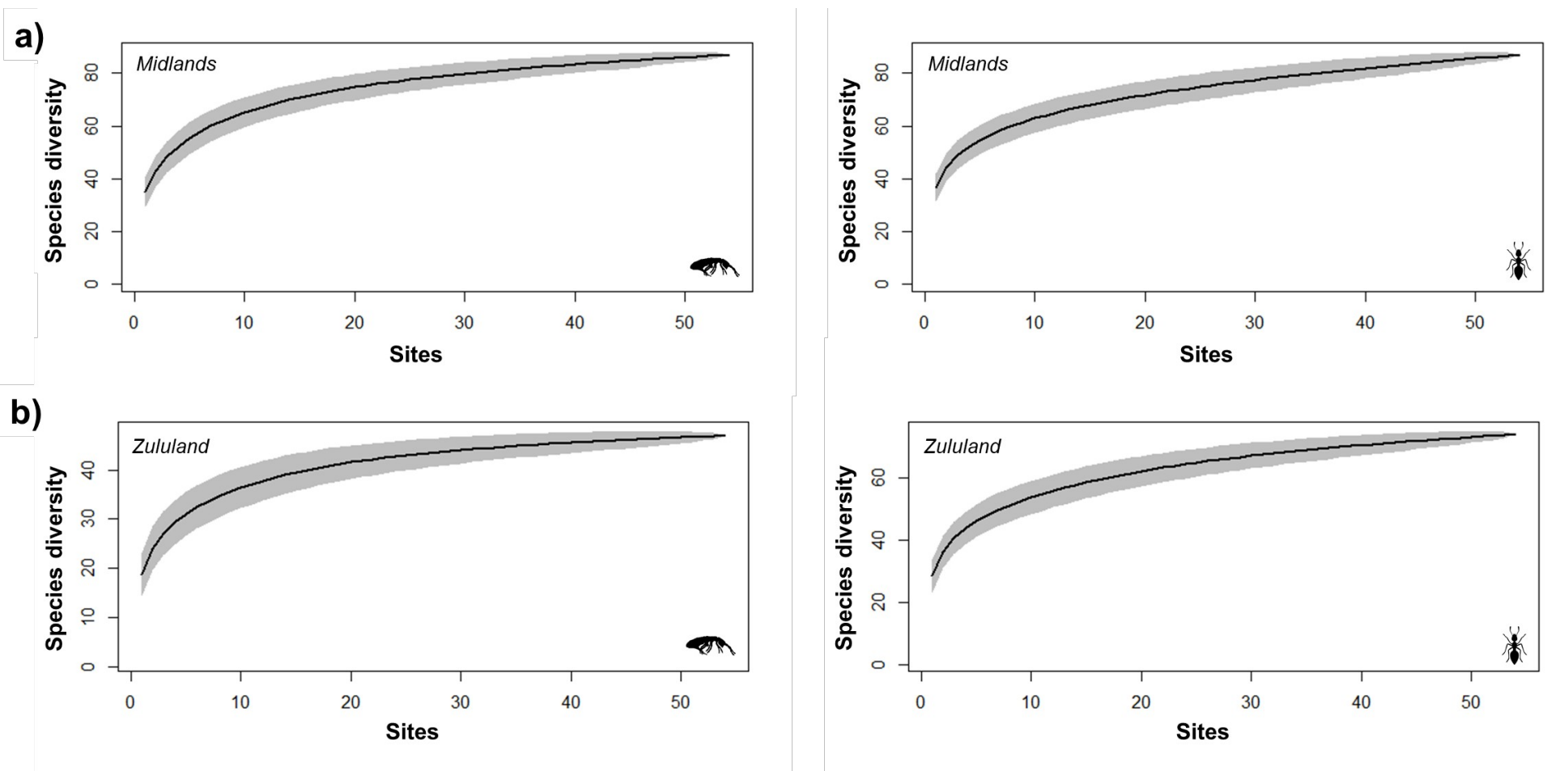


Table S7 Number of collected species (N) and species estimators Chao2 and Jackknife2 for ants (Formicidae) and springtails (Collembola) within each biotope type in the Midlands and Zululand.

	Formicidae			Collembola		
	N	Chao2	Jackknife2	N	Chao2	Jackknife2
Midlands						
Indigenous forests	24	30.25	27.99	50	54.65	56.00
Grasslands	75	112.56	102.99	57	65.45	72.00
Eucalypt compartments	33	37.00	39.98	64	100.45	87.97
Zululand						
Indigenous forest	56	68.03	67.99	25	49.00	29.98
Grassland	31	37.40	32.00	35	45.08	46.98
Eucalypt compartments	39	120.00	53.97	21	31.67	22.00